

E Full results

The detailed LLM performances on ICL benchmarks are presented in tables below:

- Table 7: ARC-AGI
- Table 8: MiniARC
- Table 9: 1D-ARC
- Table 10: SCAN
- Table 11: MiniSCAN
- Table 12: COGS
- Table 13: SALT
- Table 14: List Function
- Table 15: RAVEN

Table 7: Detailed LLM Performances on ARC-AGI.

Model	Direct	CoT		ReAct		ToT	
	Acc (%)	Acc (%)	# tokens	Acc (%)	# tokens	Acc (%)	# tokens
<i>Open-source</i>							
Deepseek-V3	15.93	12.81 (-3.12)	800.33	11.50 (-4.43)	719.48	11.98 (-3.95)	1055.27
Llama3.1-8B	3.95	2.75 (-1.20)	1324.51	2.28 (-1.67)	1735.61	2.16 (-1.79)	2125.06
Llama3.1-70B	10.66	8.02 (-2.64)	645.38	10.25 (-0.41)	681.89	7.66 (-3.00)	1477.78
Llama3.1-405B	16.45	10.42 (-6.03)	699.35	11.86 (-4.59)	1434.73	8.54 (-7.91)	1147.96
Qwen2.5-7B	4.31	1.92 (-2.39)	1681.24	0.96 (-3.35)	1841.80	1.32 (-2.99)	1988.00
Qwen2.5-72B	11.98	11.14 (-0.84)	1021.80	1.80 (-10.18)	1094.05	7.90 (-4.08)	1430.62
Mistral-7B	0.36	0.48 (+0.12)	672.39	0.96 (+0.60)	758.50	0.48 (+0.12)	902.04
Mistral-Small 3	10.30	5.99 (-4.31)	1768.28	0.72 (-9.58)	409.83	5.15 (-5.15)	1619.02
<i>Proprietary</i>							
Gemini-1.5-flash	11.26	7.90 (-3.36)	727.77	12.33 (+1.07)	872.89	13.11 (+1.85)	930.87
Gemini-1.5-pro	17.25	13.41 (-3.84)	787.59	4.08 (-13.17)	1080.24	15.15 (-2.10)	840.94
Gemini-2.0-flash	14.25	10.06 (-4.19)	867.74	11.67 (-2.58)	1005.52	9.34 (-4.91)	3645.00
GPT-3.5-turbo	4.09	4.55 (+0.46)	459.40	3.29 (-0.80)	213.74	2.44 (-1.65)	255.09
GPT-4o-mini	5.51	5.15 (-0.36)	632.05	4.01 (-1.50)	754.46	3.71 (-1.80)	840.72
GPT-4o	13.77	10.42 (-3.35)	708.77	11.55 (-2.22)	777.92	8.98 (-4.79)	1019.13
Average	10.01	7.50 (-2.51)	914.04	6.23 (-3.78)	955.76	6.99 (-3.02)	1376.96

Table 8: Detailed LLM Performances on MiniARC.

Model	Direct	CoT		ReAct		ToT	
	Acc (%)	Acc (%)	# tokens	Acc (%)	# tokens	Acc (%)	# tokens
<i>Open-source</i>							
Deepseek-V3	27.52	15.44 (-12.08)	710.46	15.44 (-12.08)	789.48	14.77 (-12.75)	395.23
Gemma2-9B	11.41	3.36 (-8.05)	194.93	1.34 (-10.07)	435.62	2.68 (-8.73)	138.50
Gemma2-27B	14.09	11.41 (-2.68)	142.81	3.36 (-10.73)	332.99	6.71 (-7.38)	114.46
Llama3.1-8B	10.74	6.71 (-4.03)	566.13	4.70 (-6.04)	1537.84	3.36 (-7.38)	323.08
Llama3.1-70B	21.48	14.09 (-7.39)	173.00	13.42 (-8.06)	816.99	8.72 (-12.76)	126.50
Llama3.1-405B	26.71	16.11 (-10.60)	522.15	15.44 (-11.27)	645.18	14.09 (-12.62)	301.58
Qwen2.5-7B	10.07	2.01 (-8.06)	587.68	0.00 (-10.07)	811.81	4.03 (-6.04)	335.85
Qwen2.5-72B	23.49	7.38 (-16.11)	592.43	0.67 (-22.82)	881.40	10.07 (-13.42)	336.21
Mistral-7B	2.68	0.67 (-2.01)	173.39	2.68 (0.00)	356.87	0.67 (-2.01)	108.88
Mistral-Small 3	16.11	6.71 (-9.40)	805.96	2.01 (-14.10)	750.36	9.40 (-6.71)	445.04
<i>Proprietary</i>							
Gemini-1.5-flash	16.11	14.09 (-2.02)	278.79	10.74 (-5.37)	592.56	10.07 (-6.04)	181.90
Gemini-1.5-pro	24.83	21.48 (-3.35)	626.63	15.44 (-9.39)	460.40	16.78 (-8.05)	205.33
Gemini-2.0-flash	25.50	14.09 (-11.41)	626.63	18.12 (-7.38)	1048.62	11.41 (-14.09)	246.71
GPT-3.5-turbo	7.38	5.37 (-2.01)	153.30	4.70 (-2.68)	193.79	6.04 (-1.34)	116.19
GPT-4o-mini	13.42	10.74 (-2.68)	252.89	11.41 (-2.01)	407.54	8.05 (-5.37)	165.94
GPT-4o	22.15	16.11 (-6.04)	308.75	19.61 (-2.54)	552.40	14.77 (-7.38)	194.06
Average	17.11	10.36 (-6.75)	419.75	8.69 (-8.42)	663.37	8.85 (-8.26)	233.47

Table 9: Detailed LLM Performances on 1D-ARC.

Model	Direct	CoT		ReAct		ToT	
	Acc (%)	Acc (%)	# tokens	Acc (%)	# tokens	Acc (%)	# tokens
<i>Open-source</i>							
Deepseek-V3	69.70	66.26 (-3.44)	723.92	66.93 (-2.77)	775.21	55.38 (-14.32)	733.13
Gemma2-9B	28.30	20.20 (-8.10)	171.04	11.54 (-16.76)	239.97	11.10 (-17.20)	357.99
Gemma2-27B	41.62	31.08 (-10.54)	141.58	22.42 (-19.20)	212.37	23.75 (-17.87)	287.10
Llama3.1-8B	23.20	13.21 (-9.99)	484.33	12.32 (-10.88)	873.66	10.65 (-12.55)	1349.24
Llama3.1-70B	53.89	46.00 (-7.89)	165.14	40.51 (-13.38)	301.79	33.96 (-19.93)	697.90
Llama3.1-405B	60.60	58.49 (-2.11)	434.75	55.83 (-4.77)	665.89	44.28 (-16.32)	677.26
Qwen2.5-7B	25.86	13.67 (-12.19)	445.33	11.76 (-14.10)	506.17	3.88 (-21.98)	705.43
Qwen2.5-72B	51.67	48.22 (-3.45)	294.11	13.32 (-38.35)	353.90	36.40 (-15.27)	914.23
Mistral-7B	0.00	1.00 (+1.00)	193.50	1.22 (+1.22)	334.70	1.11 (+1.11)	341.51
Mistral-Small 3	47.50	32.74 (-14.76)	870.63	0.00 (-47.50)	480.67	29.74 (-17.76)	694.24
<i>Proprietary</i>							
Gemini-1.5-flash	53.27	39.84 (-13.43)	231.44	40.51 (-12.76)	466.01	34.07 (-19.20)	538.92
Gemini-1.5-pro	67.04	58.71 (-8.33)	269.24	56.27 (-10.77)	420.13	48.95 (-18.09)	388.57
Gemini-2.0-flash	60.38	50.94 (-9.44)	644.63	48.83 (-11.55)	452.14	45.51 (-14.87)	748.79
GPT-3.5-turbo	8.66	14.43 (+5.77)	174.62	15.32 (+6.66)	142.77	11.43 (+2.77)	183.69
GPT-4o-mini	26.53	19.64 (-6.89)	227.64	17.76 (-8.77)	379.46	17.43 (-9.10)	401.00
GPT-4o	42.51	44.40 (+1.89)	281.26	41.62 (-0.89)	370.70	38.40 (-4.11)	496.21
Average	41.30	34.93 (-6.37)	359.57	28.51 (-12.79)	435.97	27.88 (-13.42)	594.70

Table 10: Detailed LLM Performances on SCAN.

Model	Direct		CoT		ReAct		ToT	
	Acc (%)	Acc (%)	# tokens	Acc (%)	# tokens	Acc (%)	# tokens	
Open-source								
Deepseek-V3	91.85	81.70 (-10.15)	168.76	81.90 (-9.95)	248.69	78.00 (-13.85)	225.57	
Gemma2-9B	32.60	38.60 (+6.00)	85.34	36.30 (+3.70)	169.07	28.17 (-4.43)	352.10	
Gemma2-27B	60.86	53.70 (-7.16)	102.49	55.30 (-5.56)	242.08	46.46 (-14.40)	430.27	
Llama3.1-8B	24.92	23.79 (-1.13)	142.52	14.19 (-10.73)	659.83	14.41 (-10.51)	1003.49	
Llama3.1-70B	68.38	60.26 (-8.12)	233.15	55.40 (-12.98)	525.03	53.70 (-14.68)	1080.48	
Llama3.1-405B	84.42	81.70 (-2.72)	154.70	86.00 (+1.58)	336.10	79.20 (-5.22)	646.67	
Qwen2.5-7B	41.32	33.53 (-7.79)	115.28	31.89 (-9.43)	174.10	31.16 (-10.16)	300.71	
Qwen2.5-72B	88.05	88.55 (+0.50)	99.52	89.48 (+1.43)	204.55	87.00 (-1.05)	180.86	
Mistral-7B	21.21	17.07 (-4.14)	122.67	20.90 (-0.31)	163.15	11.01 (-10.20)	270.43	
Mistral-Small 3	70.90	67.75 (-3.15)	156.90	52.10 (-18.80)	335.95	43.89 (-27.01)	500.01	
Proprietary								
Gemini-1.5-flash	69.30	59.80 (-9.50)	113.15	56.60 (-12.70)	236.42	46.50 (-22.80)	342.88	
Gemini-1.5-pro	93.10	87.20 (-5.90)	155.83	87.00 (-6.10)	225.16	71.00 (-22.10)	228.23	
Gemini-2.0-flash	86.30	86.20 (-0.10)	161.69	81.30 (-5.00)	355.34	76.70 (-9.60)	951.81	
GPT-3.5-turbo	37.60	41.00 (+3.40)	77.67	28.50 (-9.10)	73.53	22.80 (-14.80)	198.80	
GPT-4o-mini	51.00	57.40 (+6.40)	115.21	57.20 (+6.20)	162.19	48.20 (-2.80)	263.84	
GPT-4o	82.90	82.40 (-0.50)	147.24	83.60 (+0.70)	211.43	82.70 (-0.20)	310.11	
Average	62.79	60.04 (-2.75)	134.51	57.35 (-5.44)	270.16	51.31 (-11.48)	455.39	

Table 11: Detailed LLM Performances on MiniSCAN.

Model	Direct	CoT		ReAct		ToT	
	Acc (%)	Acc (%)	# tokens	Acc (%)	# tokens	Acc (%)	# tokens
Open-source							
Deepseek-V3	16.50	16.30 (-0.20)	252.12	18.80 (+2.30)	285.60	20.80 (+4.30)	258.93
Gemma2-9B	9.50	3.01 (-6.49)	149.56	1.00 (-8.50)	190.38	0.80 (-8.70)	403.07
Gemma2-27B	16.40	13.50 (-2.90)	107.12	8.10 (-8.30)	239.84	7.90 (-8.50)	485.17
Llama3.1-8B	1.10	2.30 (+1.20)	289.07	0.47 (-0.63)	737.72	2.60 (+1.50)	1226.43
Llama3.1-70B	27.00	19.80 (-7.20)	365.74	17.71 (-9.29)	345.85	16.70 (-10.30)	1067.96
Llama3.1-405B	33.30	32.00 (-1.30)	383.64	35.00 (+1.70)	385.51	24.60 (-8.70)	741.70
Qwen2.5-7B	2.80	1.42 (-1.38)	146.22	1.34 (-1.46)	182.02	3.77 (+0.97)	881.11
Qwen2.5-72B	20.00	22.00 (+2.00)	204.60	20.70 (+0.70)	261.85	19.50 (-0.50)	316.75
Mistral-7B	0.20	1.00 (+0.80)	182.66	1.13 (+0.93)	198.82	0.58 (+0.38)	348.98
Mistral-Small 3	29.63	28.30 (-1.33)	385.82	23.50 (-6.13)	297.09	30.59 (+0.96)	575.90
Proprietary							
Gemini-1.5-flash	35.40	32.10 (-3.30)	233.42	28.00 (-7.40)	523.10	23.00 (-12.40)	573.68
Gemini-1.5-pro	46.80	45.75 (-1.05)	299.28	42.60 (-4.20)	523.00	41.00 (-5.80)	258.07
Gemini-2.0-flash	47.30	31.30 (-16.00)	329.51	31.20 (-16.10)	523.10	32.10 (-15.20)	745.29
GPT-3.5-turbo	6.10	1.40 (-4.70)	98.67	0.30 (-5.80)	103.16	1.50 (-4.60)	261.70
GPT-4o-mini	6.30	4.10 (-2.20)	160.05	2.80 (-3.50)	208.32	1.60 (-4.70)	329.40
GPT-4o	35.80	20.60 (-15.20)	252.39	18.90 (-16.90)	276.85	19.60 (-16.20)	399.70
Average	20.88	17.18 (-3.70)	239.99	15.72 (-5.16)	330.14	15.42 (-5.46)	554.62